

Precision in Algorithms

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can translate a short paper calculation into a Python script.

I can predict and explain the order of executed statements.

I can keep expected and observed output separate when checking a program.

Remember before reading

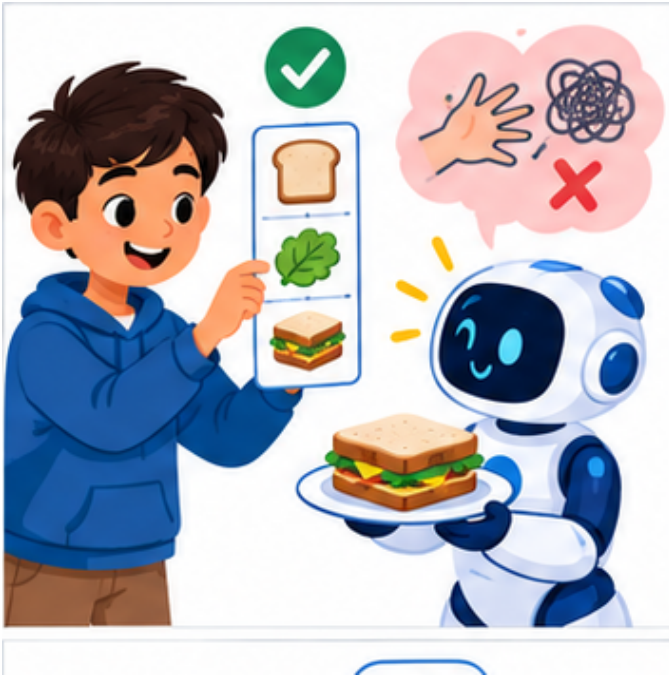
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Expo and Defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 1: Precision in Algorithms

Computers follow stated steps, not intended meaning.



THINKER MISSION

Write two sandwich algorithms; expose hidden assumptions.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Four packs contain six counters each. What steps find and show the total?

This week's end goal

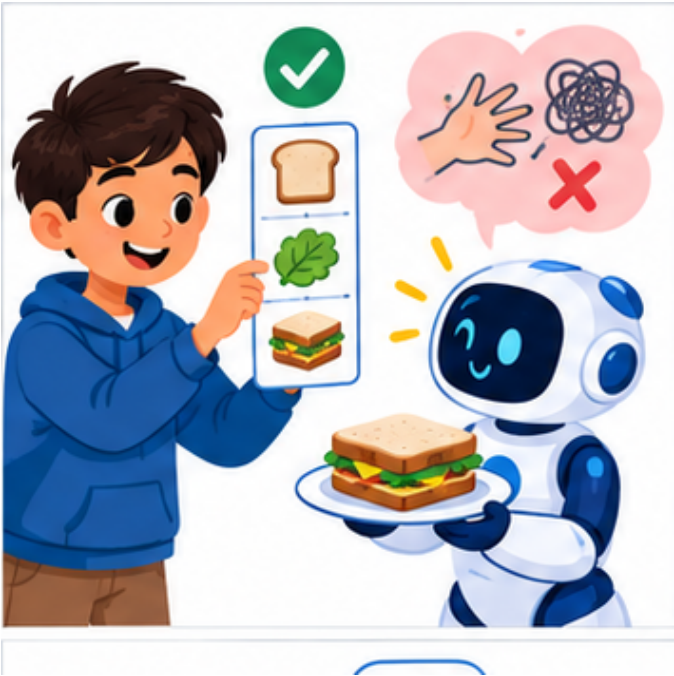
By the end, I can translate an ordered algorithm into Python and check state after each line.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use precision in algorithms to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

In Grade 6 you traced rules on paper. Now you will write a small Python program and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser. The requirement is to find the total counters in equal packs and display that total. For this first program, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed input.

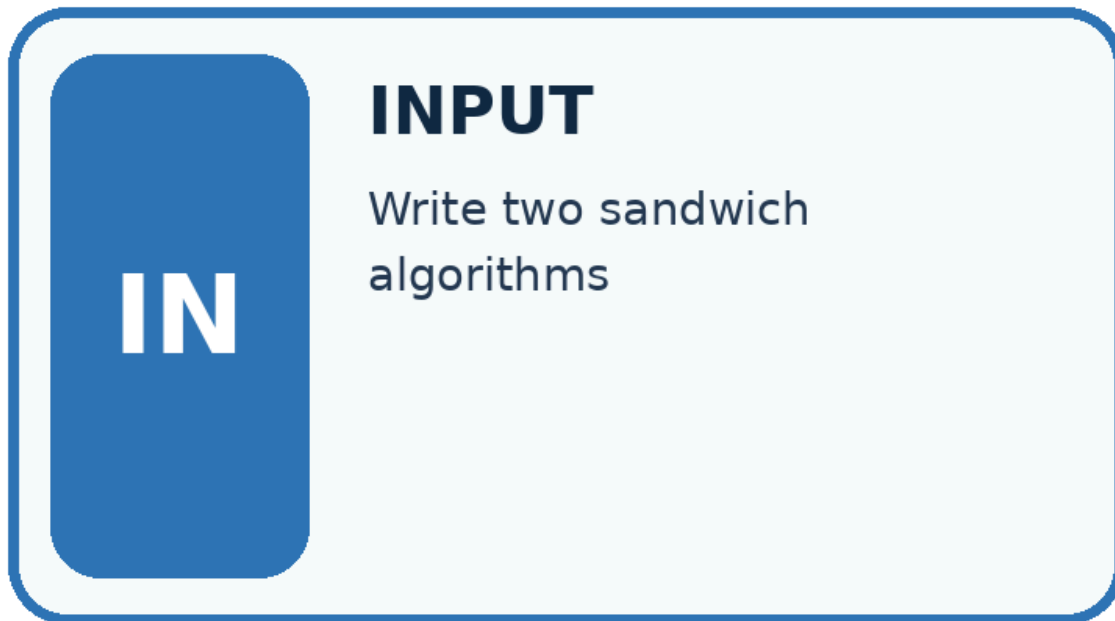
Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor's Run command. Read the output area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Computers follow stated steps, not intended meaning.





OUTPUT

expose hidden assumptions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A calculation from paper to output

Paper step · Python instruction · Effect

Paper step: Record pack count

Python instruction: `packs = 4`

Effect: Name holds 4

Paper step: Multiply two quantities

Python instruction: `total = packs * per_pack`

Effect: Name holds product

Paper step: Show the answer

Python instruction: `print(total)`

Effect: Displays the value

Assign `per_pack` before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

```
packs = 4

per_pack = 6

total = packs * per_pack

print(total)
```

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. `print(total)` displays the value of `total`; it does not display the letters in the name. Type ordinary straight characters and keep the parentheses. The Python prompt symbols `>>>` are not part of this saved script.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL

Expected everyday case



BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Set boxes=0, keeping the two loose books.

Prediction: The printed result is 2.

Reason: $0 \times 5 + 2 = 2$; there is no rule saying zero boxes means zero loose books.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

PRECISE

clear enough to repeat

3

ASSUMPTION

an unstated idea treated as true

4

TEST CASE

one input used to check behavior

Word · My own example · Not this / boundary

Word: algorithm

My own example: A specified sequence of steps for performing a task or solving a class of problems.

Not this / boundary: _____

Word: precise

My own example: Specific enough that a reader can distinguish the intended meaning or reproduce the action.

Not this / boundary: _____

Word: assumption

My own example: Something treated as true for reasoning that has not been established by the supplied evidence.

Not this / boundary: _____

Word: test case

My own example: A specified input and conditions with an expected result or measurement rule.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Program

Meaning: Instructions written in a language a computer can execute

Word: Interpreter

Meaning: Software that executes Python instructions

Word: Assignment

Meaning: Binding a name to the value of an expression

Word: Output

Meaning: Information a program displays or returns

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 1 • WORKED EVIDENCE

- 1 Vague: 'Add the filling.'
- 2 Question: Which filling? How much? Where?
- 3 Precise: place 2 cucumber slices on the center of bread A.
- 4 Test with a new partner: identical results are evidence of precision.

1. Vague: 'Add the filling.'

2. Question: Which filling? How much? Where?

3. Precise: place 2 cucumber slices on the center of bread A.

4. Test with a new partner: identical results are evidence of precision.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After line · Names and values · Displayed output

After line: packs = 4

Names and values: packs is 4

Displayed output: Nothing yet

After line: per_pack = 6

Names and values: packs is 4; per_pack is 6

Displayed output: Nothing yet

After line: total = packs * per_pack

Names and values: total is 24

Displayed output: Nothing yet

After line: print(total)

Names and values: Values are unchanged

Displayed output: 24

Write 24 as the expected output before running. Then copy the actual output into a separate column. A matching result supports this case, not every possible input. A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the output record without running.

Why is there no displayed output after the third line? What changes if `print(total)` is placed first in a fresh run, before any assignment? Explain using which names exist.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Precision in Algorithms

A fictional book-box counter has three boxes with five books each and two loose books.

The facts we will use

Python instructions: `boxes=3; total=boxes*5; total=total+2; print(total)`.

Run the instructions in this order with ordinary integer arithmetic.

Precision in Algorithms

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read input

boxes is assigned 3.



2. Multiply

total becomes $3 \times 5 = 15$.



3. Update total

total becomes its current value 15 plus 2: 17.



4. Print

The program prints 17 once.

CHANGE ONE THING

Set boxes=0, keeping the two loose books.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read input

What happens and why: boxes is assigned 3.

Step: 2. Multiply

What happens and why: total becomes $3 \times 5 = 15$.

Step: 3. Update total

What happens and why: total becomes its current value 15 plus 2: 17.

Step: 4. Print

What happens and why: The program prints 17 once.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Assignment replaces a variable's stored value. In `total=total+2`, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left. The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict output before running.

What this example does not establish

These statements count supplied books; they do not validate input types or infer stock from measurements.

Read the labelled picture: If `print(total)` moves before `total=total+2`, what gets printed for the original input?

Change one thing: Set `boxes=0`, keeping the two loose books. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A longer instruction is automatically more precise. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Save the file with the input values, expected and observed outputs, and a short explanation. This program

follows a written multiplication rule. It has not trained a model, predicted somebody’s preferences or checked that real packs contain the stated number. Its conclusion depends on the supplied data. Week 2 examines the values and types behind these names.

A script runs without an error and displays 11 for five packs of six. Is it correct for the stated requirement? Give the required result, the likely wrong operation and one check after repair. State whether your evidence came from a device run or a paper trace.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change only packs to 5. Predict the result from five groups of six, save, then run the entire file again. The expected output is 30. Keep the original four-pack result beside the new one and label the changed input. A comment can explain an instruction: Python ignores text after # outside a string. A comment does not change the calculation.

```
packs = 5 # five equal packs
per_pack = 6
total = packs * per_pack
print(total)
```

A program can run successfully and still solve the wrong problem. Replacing * with + would display 11 for this input, but addition does not count five groups of six. Use the requirement and a paper calculation to check meaning; the absence of an error message is not proof of correctness.

Practise with less help: Predict then observe

Use workbook W2 for the connected example “From a paper algorithm to Python”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for From a paper algorithm to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

If `print(total)` moves before `total=total+2`, what gets printed for the original input?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Trade sandwich instructions, follow them literally, mark every place where two interpretations are possible, then repair one cause at a time.

Trade sandwich instructions, follow them literally, mark every place where two interpretations are possible, then repair one cause at a time.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Write exact scoring steps so two people rank the same book the same way.

THIS WEEK'S CHECKPOINT Write exact scoring steps so two people rank the same book the same way.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

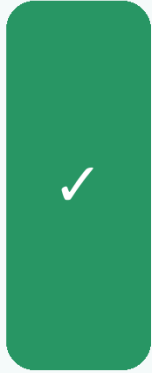
Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Write two sandwich algorithms; expose hidden assumptions.

CHALLENGE Write two sandwich algorithms; expose hidden assumptions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A sandwich algorithm says add spread, close it, cut it, without stating materials or directions.

The question

Expose assumptions and write a precise paper simulation.

Worked explanation

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Make hidden instructions explicit

Evidence or stage · Value or result

Evidence or stage: Materials

Value or result: Two paper bread squares

Evidence or stage: Spread

Value or result: Mark one spoon-sized area

Evidence or stage: Close

Value or result: Place second square on top

Evidence or stage: Cut

Value or result: Draw a diagonal line

A sandwich algorithm says add spread, close it, cut it, without stating materials or directions.

Expose assumptions and write a precise paper simulation.

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

14. Try a new case independently

Independent case: Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its output. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write exact scoring steps so two people rank the same book the same way.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Write an algorithm for borrowing a classroom book when the book may already be on loan. Include inputs, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional book-box counter has three boxes with five books each and two loose books.

By the end, I can · How I will show it

By the end, I can: Translate a short paper calculation into a Python script

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Predict and explain the order of executed statements

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Keep expected and observed output separate when checking a program

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Translate a short paper calculation into a Python script

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-box counter has three boxes with five books each and two loose books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The program prints 17 once.

1. Read input
2. Print
3. Multiply
4. Update total

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Print The program prints 17 once. Assignment replaces a variable's stored value. In $\text{total} = \text{total} + 2$, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left. The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict output before running.

Practice 2: explain the deciding evidence

Changed-case practice: Set $\text{boxes} = 0$, keeping the two loose books. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The printed result is 2.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The printed result is 2. $0 \times 5 + 2 = 2$; there is no rule saying zero boxes means zero loose books.

Repair a missing building block

Review Expo and Defense (ai-1/36). Goal: Evaluate a saved candidate and a manual baseline fairly. Short practice: Write one defensible success statement and one claim the evidence cannot support.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.